


24671 Electromechanical Systems Design

# Multirotor-Operated Lab Experiments (M.O.L.E.)

Midsemester Design Report

**Date:** ¡Month Day, Year¿
**Group:** Group 3: M.O.L.E

| **Team member** | **Signature** |
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## Executive Summary

**Problem.** *[In one paragraph, state the user problem, operating context, and why it matters.]*

**Product and conceptual design.** *[Summarize the proposed system, its main features, and the design concept selected.]*

**Final prototype.** *[State exactly what the final prototype will demonstrate, and distinguish demonstrated functionality from deferred features.]*

**Results.** *[Summarize the prototype tests, the most important quantitative results, and the principal design decisions that follow from them.]*

**Takeaway.** *[Write this section last. It should stand alone as an abstract of the full report.]*

# 1 Problem Definition

## 1.1 Problem Description

*[Describe the problem in measurable terms. Explain the current workflow, the users affected, the environment in which the system operates, and the consequences of not solving the problem. Define any technical terms.]*

## 1.2 Markets Addressed

*[Identify the market or application segments, expected users, purchasers, and deployment environments. Explain the initial beachhead market and possible future markets.]*

## 1.3 Primary Competitors and Existing Solutions

*[Compare direct competitors, adjacent products, and the current manual or improvised solution. Use consistent comparison criteria.]*

Table 1: Preliminary competitor comparison.

| Competitor/solution | Core offering | Strengths | Limitations | Web link |
|---|---|---|---|---|
| *[Competitor 1]* | *[...]* | *[...]* | *[...]* | https://example.com |
| *[Competitor 2]* | *[...]* | *[...]* | *[...]* | https://example.com |
| *[Current practice]* | *[...]* | *[...]* | *[...]* | *[N/A]* |

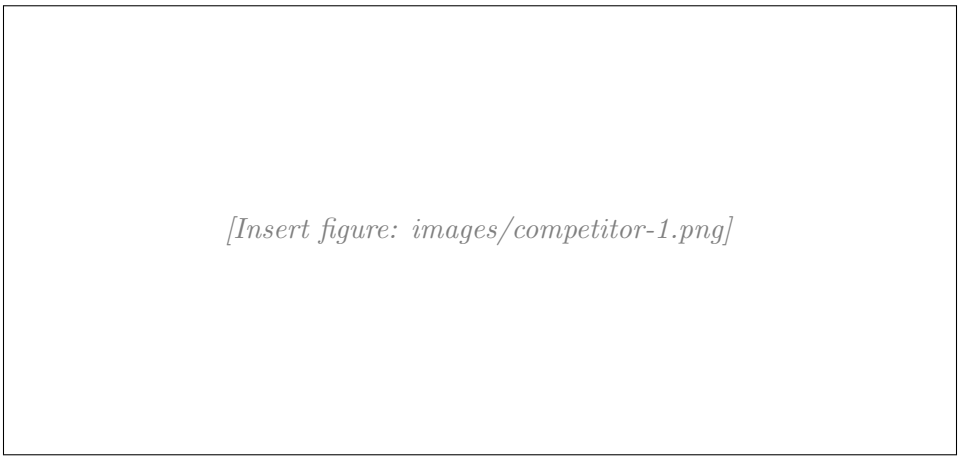



Figure 1: Key competitor or existing solution. Replace the filename and cite the source.

### 1.4 Assumptions and Constraints

- *[Technical assumption, e.g., available power, network, or lab infrastructure.]*
- *[User or operational assumption.]*
- *[Safety, budget, schedule, size, weight, or regulatory constraint.]*
- *[What is explicitly out of scope for this project?]*

## 2 Stakeholders and Customer Needs

### 2.1 Stakeholder Identification

*[List primary users, customers, operators, maintainers, instructors, administrators, suppliers, regulators, and other parties affected by the design. Explain each stakeholder's influence and interest.]*

Table 2: Stakeholder map.

| Stakeholder | Role/context | Goals | Influence/priority |
|---|---|---|---|
| *[Primary user]* | *[...]* | *[...]* | *[High]* |
| *[Customer/purchaser]* | *[...]* | *[...]* | *[High]* |
| *[Maintainer]* | *[...]* | *[...]* | *[Medium]* |

### 2.2 External Outreach

*[Describe interviews, surveys, observations, expert consultations, or other outreach. State who participated, when, how many responses were obtained, and how the evidence was analyzed.]*

### 2.3 Internal Need Generation

*[Describe needs generated from team brainstorming, engineering analysis, course requirements, and lessons from early prototypes.]*

### 2.4 Organization and Classification

*[Explain how raw statements were converted into needs, normalized, de-duplicated, and organized by user, function, or priority.]*

### 2.5 Final Customer Needs

Table 3: Final customer needs.

| ID | Need statement | Source | Priority |
|---|---|---|---|
| *[N1]* | *[The system should...]* | *[Interview/survey]* | *[1–5]* |
| *[N2]* | *[The system should...]* | *[Observation]* | *[1–5]* |
| *[N3]* | *[The system should...]* | *[Team analysis]* | *[1–5]* |

# 3 Target Specifications

## 3.1 Mapping Needs to Metrics

*[Explain how each important need was translated into a measurable engineering metric, including units, direction of improvement, and test method.]*

Table 4: Need-to-metric mapping.

| Need ID | Metric | Units | Why it matters | Verification method |
|---|---|---|---|---|
| *[N1]* | *[Response time]* | s | *[...]* | *[Bench test]* |
| *[N2]* | *[Accuracy]* | % | *[...]* | *[Experiment]* |

## 3.2 Competitive Analysis

*[Compare competitor performance against the metrics above. Identify gaps and opportunities rather than only listing features.]*

## 3.3 Target Specifications

Table 5: Final target specifications.

| Metric | Minimum | Target | Ideal | Test method |
|---|---|---|---|---|
| *[Metric 1]* | *[...]* | *[...]* | *[...]* | *[...]* |
| *[Metric 2]* | *[...]* | *[...]* | *[...]* | *[...]* |
| *[Metric 3]* | *[...]* | *[...]* | *[...]* | *[...]* |

# 4 Concept Generation

## 4.1 Functional Decomposition

*[Define the overall system function and decompose it into subfunctions. A block diagram should show inputs, outputs, flows, and system boundaries.]*

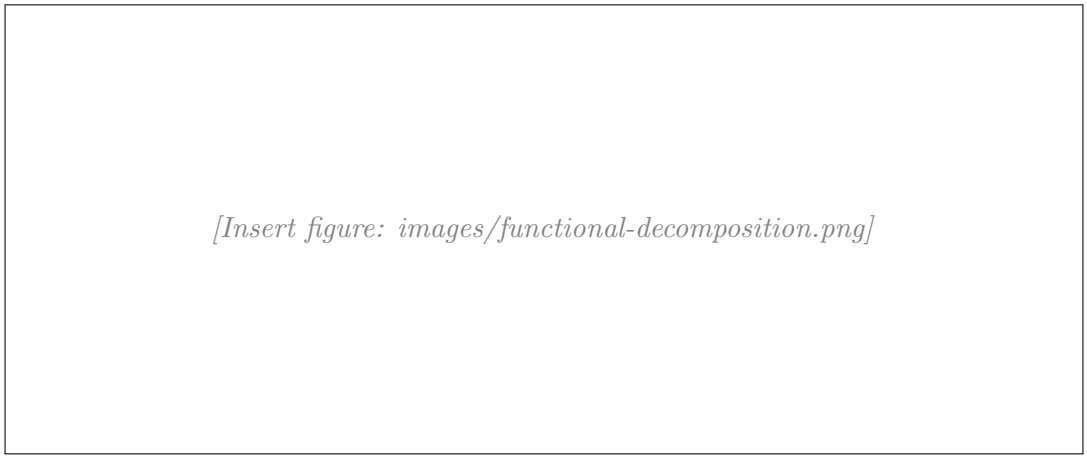



Figure 2: Functional decomposition of the proposed system.

## 4.2 Concept Generation Activities

### 4.2.1 External Search

*[Describe literature, patent, product, standards, and web searches. Record the sources and explain what design principles were adopted or rejected.]*

### 4.2.2 Internal Search

*[Describe brainstorming, analogous products, morphological charts, sketching, workshops, and other team activities.]*

## 4.3 Subsystem Concepts

*[Present the alternatives generated for each major subsystem. Move additional sketches to Appendix A.]*

Table 6: Morphological chart of subsystem alternatives.

| Subsystem | Alternative A | Alternative B | Alternative C |
|---|---|---|---|
| *[Subsystem 1]* | *[...]* | *[...]* | *[...]* |
| *[Subsystem 2]* | *[...]* | *[...]* | *[...]* |
| *[Subsystem 3]* | *[...]* | *[...]* | *[...]* |

### 4.4 Full System Concepts

*[Show the full-system combinations considered from the subsystem alternatives. Give each concept a short name and describe its operating principle, advantages, risks, and unresolved questions.]*

*[Insert figure: images/system-concepts.png]*

Figure 3: Overview of full-system concepts considered.

## 5 Concept Selection

### 5.1 Selection Process

*[Describe screening, feasibility checks, a Pugh matrix, weighted decision matrix, or other methods. Explain criteria, weights, scoring scale, and how disagreements were resolved.]*

Table 7: Weighted concept-selection matrix.

| Criterion (weight) | Concept A | Concept B | Concept C | Rationale |
|---|---|---|---|---|
| *[Criterion 1 (0.25)]* | *[score]* | *[score]* | *[score]* | *[...]* |
| *[Criterion 2 (0.25)]* | *[score]* | *[score]* | *[score]* | *[...]* |
| *[Criterion 3 (0.25)]* | *[score]* | *[score]* | *[score]* | *[...]* |
| *[Criterion 4 (0.25)]* | *[score]* | *[score]* | *[score]* | *[...]* |
| **Weighted total** | *[...]* | *[...]* | *[...]* | |

### 5.2 Top Concept(s): Assessment and Justification

*[Justify the selected concept using evidence, not only preference. Discuss tradeoffs, risks, expected performance, cost, manufacturability, and alignment with customer needs.]*

### 5.3 Detailed Model of Top Concept(s)

*[Include CAD, architecture diagrams, circuit diagrams, software models, interface descriptions, and key dimensions.]*

*[Insert figure: images/top-concept-cad.png]*

Figure 4: Detailed model of the selected concept.

## 6 Prototype Summary

### 6.1 Prototype Design

#### 6.1.1 Questions to Answer

1. *[Question about technical feasibility.]*

2. *[Question about user interaction or workflow.]*
3. *[Question about performance against a target specification.]*
4. *[Question about risk, reliability, or safety.]*

### 6.1.2 Design and Construction Process

*[Describe the prototype architecture, materials, electronics, software, fabrication, assembly, and integration. Explain the sequence of decisions and document major iterations. Include photographs with captions.]*

*[Insert figure: images/prototype-overview.jpg]*

Figure 5: Prototype overview.

*[Insert figure: images/prototype-build.jpg]*

Figure 6: Prototype during construction or integration.

## 6.2 Prototype Testing

### 6.2.1 Test Plan

*[For each experiment, state the objective, setup, independent and dependent variables, controls, sample size, procedure, and pass/fail criterion.]*

Table 8: Prototype test plan.

| Test | Question | Measured output | Pass criterion | Status |
|---|---|---|---|---|
| *[T1]* | *[...]* | *[...]* | *[...]* | *[Planned/complete]* |
| *[T2]* | *[...]* | *[...]* | *[...]* | *[Planned/complete]* |

### 6.2.2 Data Analysis

*[Present raw-data processing, uncertainty, summary statistics, comparisons to targets, and limitations. Do not show only successful trials.]*

*[Insert figure: images/test-results.png]*

Figure 7: Prototype test results. Replace with a plot generated from the measured data.

Table 9: Summary of prototype test results.

| Metric | Target | Measured result | Interpretation / next action |
|---|---|---|---|
| *[Metric 1]* | *[...]* | *[...]* | *[...]* |
| *[Metric 2]* | *[...]* | *[...]* | *[...]* |

#### 6.2.3 Design Implications

*[Explain how the test results changed the next design iteration, specifications, concept choice, risk register, or test plan.]*

## 7 Project Planning

### 7.1 Schedule and Gantt Chart

*[Summarize completed work, current work, dependencies, milestones, and schedule risks. Include a Gantt chart exported as a high-resolution image or generated in this section.]*

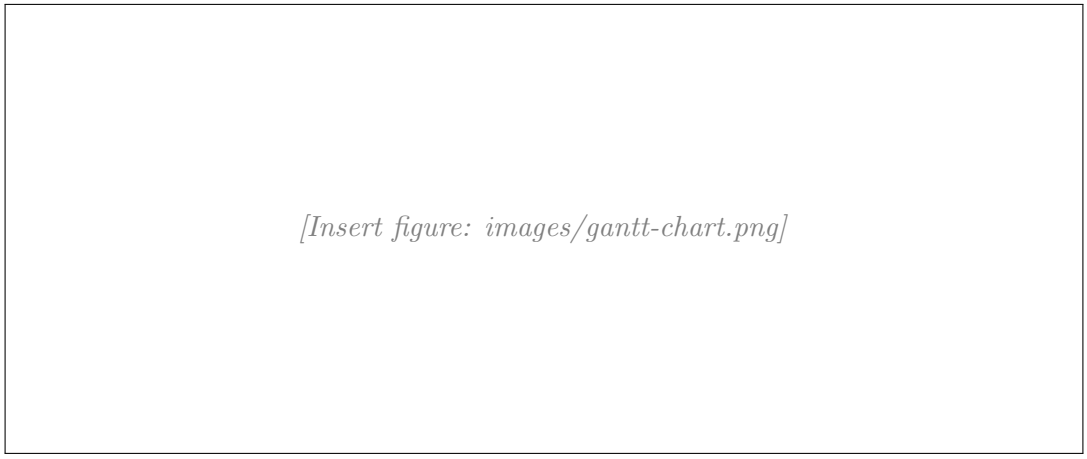



Figure 8: Project Gantt chart.

### 7.2 Prototype Planning

#### 7.2.1 Next Prototype

*[Provide a detailed build plan: components, dimensions, interfaces, fabrication steps, software work, responsible team member, cost, and expected completion date.]*

#### 7.2.2 Experiments

*[List the experiments that will be conducted on the next prototype. For each experiment, state the question, setup, data to collect, and decision rule.]*

#### 7.2.3 Expected Final Prototype Scope

*[Describe the features expected in the final prototype. Clearly separate must-have features from features deferred until a later product version. Explain why features were deferred.]*

Table 10: Planned final-prototype features.

| **Feature** | **Final prototype?** | **Evidence of completion** | **Rationale / deferral plan** |
|---|---|---|---|
| *[Feature 1]* | *[Yes/No]* | *[Test/demo]* | *[...]* |
| *[Feature 2]* | *[Yes/No]* | *[Test/demo]* | *[...]* |

## References

[1] *[Author or organization.] [Title of source or competitor page].* https://example.com, *[publication date or access date]*.

[2] *[Add course text, design-method reference, standard, paper, or other source used in the report.]*

## A Individual Concept Sketches

*[Include at least one clearly labeled page or figure for each team member's individual concepts.]*

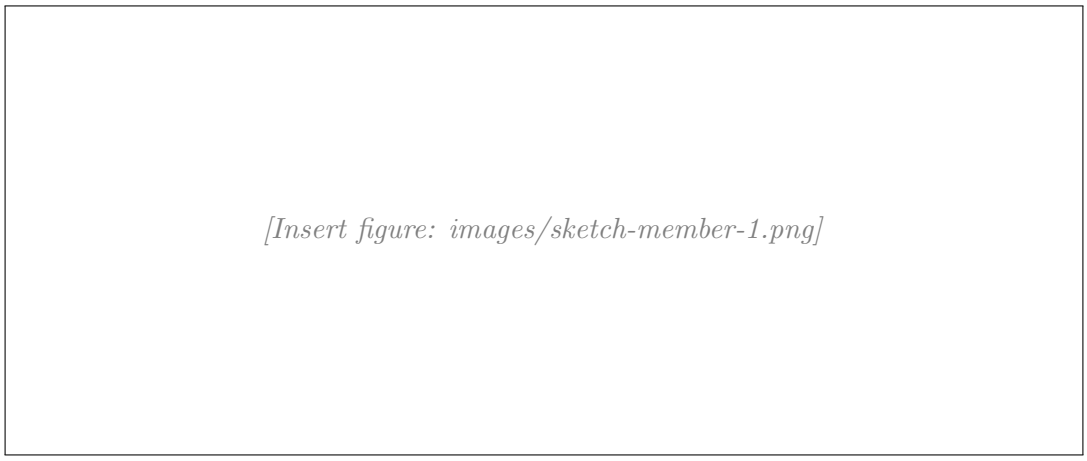



Figure 9: Concept sketches by *[Team member 1]*.

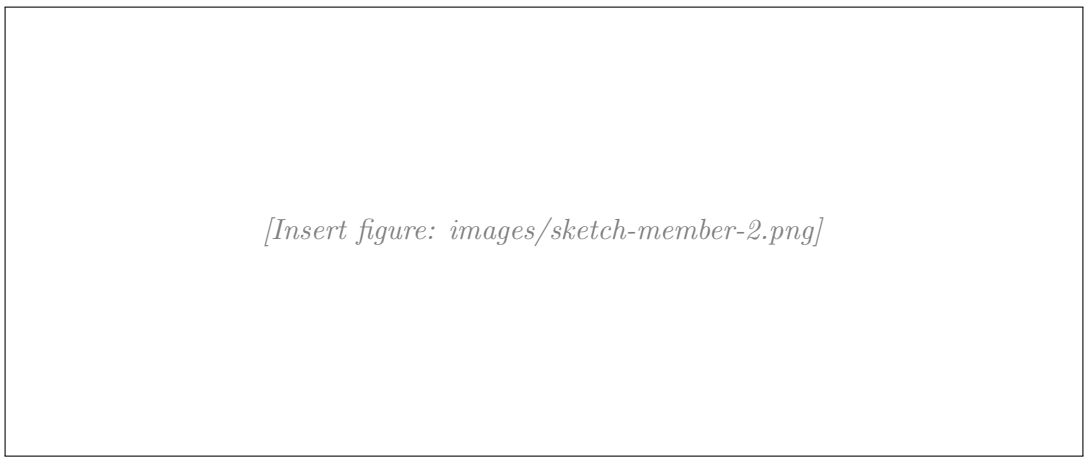



Figure 10: Concept sketches by *[Team member 2]*.

## B Survey Instrument and Interview Questions

*[Include the complete survey, interview script, consent language, and response scale. Do not include personally identifying information unless permission has been obtained.]*

1. *[Question 1]*
2. *[Question 2]*
3. *[Question 3]*

## C Work Distribution

*[Describe each team member's specific contribution to design exercises, prototyping, testing, analysis, and report writing.]*

Table 11: Work distribution.

| **Team member** | **Design/prototyping contribution** | **Report contribution** |
|---|---|---|
| *[Name 1]* | *[Specific tasks]* | *[Sections, figures, analysis, editing]* |
| *[Name 2]* | *[Specific tasks]* | *[Sections, figures, analysis, editing]* |
| *[Name 3]* | *[Specific tasks]* | *[Sections, figures, analysis, editing]* |
| *[Name 4]* | *[Specific tasks]* | *[Sections, figures, analysis, editing]* |

## D Additional Supporting Material

*[Add detailed calculations, complete raw data, risk register, additional concept sketches, CAD drawings, code excerpts, or other material that supports the report but would interrupt the main narrative.]*